Spring Boot Roadmap

October → December 2026 | Beginner → Intermediate → Advanced

■ Goal

Build a strong Spring Boot foundation, create production-style REST APIs, learn security and testing, and finish with advanced Spring Boot and Microservices.

■ October — Beginner → Spring Boot Fundamentals

Week 1: Java Revision + Spring Basics

- Java OOP: Classes & Objects, Encapsulation, Inheritance, Polymorphism, Abstraction
- Interfaces, Exception Handling, Collections, Generics
- Lambda expressions, Stream API, Optional
- Maven basics: pom.xml, dependencies, lifecycle, common commands
- Spring: IoC, Dependency Injection, Bean, ApplicationContext
- Annotations: @Component, @Service, @Repository, @Controller, @Autowired, @Configuration, @Bean
- Project: Employee Management Application

Week 2: Spring Boot Fundamentals

- What is Spring Boot and how it differs from Spring
- Spring Initializr, project structure, @SpringBootApplication
- application.properties and application.yml
- Embedded Tomcat, Starters, Auto Configuration, Component Scanning
- Profiles and configuration properties
- Project: Student Management System with CRUD

Week 3: REST API

- REST, HTTP, JSON and HTTP status codes
- GET, POST, PUT, PATCH, DELETE
- @GetMapping, @PostMapping, @PutMapping, @PatchMapping, @DeleteMapping
- @RequestBody, @PathVariable, @RequestParam, ResponseEntity
- Build complete Student CRUD REST endpoints

Week 4: Database + JPA

- SQL: SELECT, INSERT, UPDATE, DELETE, WHERE, ORDER BY, GROUP BY, JOIN
- Primary key and foreign key
- JPA, Hibernate, Entity, JpaRepository and CRUD
- Entity relationships: @OneToOne, @OneToMany, @ManyToOne, @ManyToMany
- October project: Employee Management REST API using Spring Boot + JPA + MySQL

■ November — Intermediate Spring Boot

Week 1: DTO + Validation

- Entity vs DTO, Request DTO and Response DTO
- Mapping between Entity and DTO
- Bean Validation
- @NotNull, @NotBlank, @NotEmpty, @Size, @Email, @Min, @Max, @Pattern

Week 2: Exception Handling

- Custom exceptions and meaningful error responses
- @ExceptionHandler and @ControllerAdvice
- Global exception handling
- Standard error response: status, message and timestamp

Week 3: Spring Security

- Authentication vs Authorization
- Spring Security architecture
- Password hashing and BCrypt
- UserDetails, UserDetailsService, SecurityFilterChain
- Roles and permissions
- Build secured APIs with USER and ADMIN roles

Week 4: JWT + API Security

- JWT authentication flow
- Access tokens, refresh tokens and expiration
- Stateless authentication
- CORS and CSRF basics
- November project: Secure E-Commerce REST API
- Features: Register, Login, JWT, Products, Categories, Users, Orders and Cart

■ December — Advanced Spring Boot

Week 1: Testing

- JUnit and Mockito
- Unit testing and integration testing
- @SpringBootTest, @WebMvcTest and MockMvc
- Controller, Service, Repository and REST API testing

Week 2: Advanced Database + JPA

- Lazy vs Eager Loading
- N+1 problem
- JPQL and Native Queries
- Pagination and Sorting
- Specifications
- Transactions and @Transactional
- Database indexing and query optimization

Week 3: Production-Level Spring Boot

- SLF4J, Logback and log levels
- Profiles, environment variables and secrets
- Spring Boot Actuator: Health and Metrics
- Swagger / OpenAPI
- Caching and Redis basics
- Connection pooling and async processing
- Docker fundamentals

Week 4: Microservices

- Monolith vs Microservices
- Service boundaries and REST communication
- Spring Cloud
- Eureka Service Discovery
- API Gateway
- Config Server
- OpenFeign
- Circuit Breaker and Resilience4j
- Final project: E-Commerce Microservices

■ Final Projects

Level	Project	Main Skills
Beginner	Employee Management API	Spring Boot, REST, JPA, MySQL, CRUD
Intermediate	Secure E-Commerce API	DTO, Validation, Exceptions, Spring Security, JWT, MySQL
Advanced	E-Commerce Microservices	Microservices, Spring Cloud, Gateway, Eureka, Feign, Redis, Docker

■ Exact Learning Order

1. Java OOP → 2. Collections + Streams → 3. Maven → 4. Spring Core

5. IoC + DI → 6. Spring Boot → 7. REST API → 8. MySQL

9. JPA / Hibernate → 10. DTO → 11. Validation → 12. Exception Handling

13. Spring Security → 14. JWT → 15. Testing → 16. Advanced JPA

17. Redis → 18. Actuator → 19. Docker → 20. Microservices → 21. Spring Cloud

■ December 31 Target

- Be able to build a complete Spring Boot CRUD REST API from scratch.
- Understand JPA/Hibernate, DTOs, validation and global exception handling.
- Implement authentication and authorization using Spring Security + JWT.
- Write unit and integration tests.
- Understand production concepts such as logging, Actuator, caching and Docker.
- Understand the architecture and core components of Spring Cloud Microservices.
- Have 3 portfolio-ready projects on GitHub.

Important: Do not rush into Microservices. First become comfortable building one complete Spring Boot application end-to-end. That foundation will make Microservices much easier.